IrrCollada
Particle Systems Specification

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Notes

Particle systems are not part of the Collada specification. As a consequence, we had to wrap any particle system specific information in extra elements, using the IRRLICHT profile. For now, we are only exporting particle emitters. Forces acting on those emitters will be added later on.

Unless otherwise noted, all time values are in seconds and all angle values are in degrees.

Look at the examples of elements library_emitters**,** instance_emitter **and** emitter **for a general idea of how emitters are exported to file.**

This document makes references to elements of the COLLADA 1.4.1 specification (http://www.khronos.org/collada/) and assumes the reader is familiar with it.

Content Types

'Content Type' section was added for relevant elements. It describes the type contained by the element.

Type	Description
float	Floating point value
float3	Three floats separated by a space
int	Integer value
bool	Either 'true' or 'false'
URI	Uniform Resource Identifier
string	A string of characters

Variance Equations

These have been added in relevant sections. However, they are for indicative purposes only, as those equations were used in the original PCloud particle system.

Animated Values

Unless otherwise noted, if an exported parameter is animated in Max, corresponding COLLADA animations will be exported as well.

Materials

Materials applied to particle emitter instances are exported the same way as geometry instances (using the **bind_material** element).

Percentages

For percentage values, the convention used is that 1 == 100%.

library_emitters

Introduction

Declares a module of <emitter> elements.

Concepts

As data sets become larger and more complex, they become harder to manipulate within a single container. One approach to manage this complexity is to divide the data into smaller pieces organized by some criteria. These modular pieces can then be stored in separate resources as libraries.

Attributes

The <library_emitters> element has the following attributes:

id	**xs:ID**	A text string containing the unique identifier of the <library_emitters> element. This value must be unique within the instance document. Optional.
name	**xs:NCName**	The text string name of this element. Optional.

Related Elements

The <library_emitters> element relates to the following elements:

Parent elements	**technique**
Child	See Child Elements subsection.
Other	None

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<emitter>	See main entry.	N/A	1 or more

Details

This library is always wrapped in an extra.

Example

Here is an example of a <library_emitters> element:

```
<COLLADA>
 ...
 <scene>
  ...
 </scene>
```

```
  <extra type="libraries">
    <technique profile="IRRLICHT">
      <library_emitters>
        <emitter id="IrrPCloud01-emitter" name="IrrPCloud01">
          ...
        </emitter>
      </library_emitters>
    </technique>
  </extra>
</COLLADA>
```

instance_emitter

Introduction

Describes a particle emitter in a scene.

Concepts

The <instance_emitter> element instantiates an object described by a <emitter> element. An emitter is instantiated within the local coordinate system of its parent **<node>** or **<shape>** and that determines its shape's position, orientation, and scale.

Attributes

The <instance_emitter> element has the following attributes:

id	**xs:ID**	A text string containing the unique identifier of the <instance_emitter> element. This value must be unique within the instance document. Optional.
name	**xs:NCName**	The text string name of this element. Optional.
url	**xs:anyURI**	The URL of the location of the <emitter> element to instantiate. Required. Can refer to a local instance or external reference.

For a local instance, this is a relative URI fragment identifier that begins with the "#" character. The fragment identifier is an XPointer shorthand pointer that consists of the ID of the element to instantiate.

For an external reference, this is an absolute or relative URL.

Related Elements

The <instance_emitter> element relates to the following elements:

Parent elements	**technique**
Child	See Child Elements subsection.
Other	emitter

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<bind_material>	Binds material symbols to material instances. This allows a single emitter to be instantiated into a scene multiple times each with a different appearance. See main entry.	None	0 or 1
<extra>	Provide's arbitrary information about or related to the <instance_emitter> element. See main entry.	N/A	1

Details

Example

Here is are examples of <instance_emitter> elements:

If exported as Z-Up :

```
<COLLADA>
 ...
 <library_visual_scenes>
  <visual_scene>
   <node id="IrrPCloud01-node" name="IrrPCloud01" type="NODE">
    <translate>106.173 41.2601 814.624</translate>
    <rotate>0 0 0 0</rotate>
    <scale>1 1 1</scale>
    <extra type="instances">
     <technique profile="IRRLICHT">
      <instance_emitter url="#IrrPCloud01-emitter">
       <bind_material>
        <technique_common>
         <instance_material symbol="ColorMaterial" target="#ColorMaterial_E3999900"/>
        </technique_common>
       </bind_material>
       <extra>
        <technique profile="IRRLICHT">
         <draw_layer>0</draw_layer>
        </technique>
       </extra>
      </instance_emitter>
     </technique>
    </extra>
   </node>
  </visual_scene>
 </library_visual_scenes>
 ...
</COLLADA>
```

If exported as Y-Up :

```
<COLLADA>
 ...
 <library_visual_scenes>
  <visual_scene>
   <node id="IrrPCloud01-node" name="IrrPCloud01" type="NODE">
    <translate>106.173 41.2601 814.624</translate>
    <rotate>0 0 0 0</rotate>
    <scale>1 1 1</scale>
    <node id="IrrPCloud01-node_pivot" name="IrrPCloud01_pivot" type="NODE">
     <rotate>1 0 0 -90</rotate>
     <extra type="instances">
      <technique profile="IRRLICHT">
       <instance_emitter url="#IrrPCloud01-emitter">
        <bind_material>
         <technique_common>
          <instance_material symbol="ColorMaterial" target="#ColorMaterial_E3999900"/>
         </technique_common>
        </bind_material>
        <extra>
         <technique profile="IRRLICHT">
          <draw_layer>0</draw_layer>
         </technique>
        </extra>
       </instance_emitter>
```

```
      </technique>
     </extra>
    </node>
   </node>
  </visual_scene>
 </library_visual_scenes>
 ...
</COLLADA>
```

emitter

Introduction

Describes a particle emitter in a scene.

Attributes

The <emitter> element has the following attributes:

id	**xs:ID**	A text string containing the unique identifier of the <emitter> element. This value must be unique within the instance document. Optional.
name	**xs:NCName**	The text string name of this element. Optional.

Related Elements

The <emitter> element relates to the following elements:

Parent elements	library_emitters
Child	See Child Elements subsection.
Other	instance_emitter

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<shape>	See main entry.	N/A	1
<emission>	See main entry.	N/A	1
<randomization>	See main entry.	N/A	1
<particle_life>	See main entry.	N/A	1
<particle_size>	See main entry.	N/A	1
<particle_motion>	See main entry.	N/A	1
<particle_type>	See main entry.	N/A	1
<particle_animation_offset>	See main entry.	N/A	1
<particle_animation_scaling>	See main entry.	N/A	1
<particle_type>	See main entry.	N/A	1
<particle_spin>	See main entry.	N/A	1

Details

Example

Here is an example of a <emitter> element:

```
<emitter id="DbgIrrPCloud01-emitter" name="DbgIrrPCloud01">
 <shape sid="shape">
  <sphere sid="sphere">
   <radius>38.0419</radius>
  </sphere>
 </shape>
 <emission sid="emission">
  <max_particles>100</max_particles>
  <birth_rate sid="birth_rate">300</birth_rate>
 </emission>
 <randomization sid="randomization">
  <seed>12345</seed>
 </randomization>
 <particle_life sid="particle_life">
  <life>3.36667</life>
  <life_variance>0</life_variance>
 </particle_life>
 <particle_size sid="particle_size">
  <size>1</size>
  <size_variance sid="size_variance">0</size_variance>
  <size_grow_for>0</size_grow_for>
  <size_fade_for>0</size_fade_for>
 </particle_size>
 <particle_motion sid="particle_motion">
  <speed>0</speed>
  <speed_variance sid="speed_variance">0</speed_variance>
  <direction sid="direction">
   <variable_vector sid="variable_vector">
    <vector>1 0 -10</vector>
    <variance>0.5</variance>
   </variable_vector>
  </direction>
 </particle_motion>
 <particle_type sid="particle_type">
  <standard_particle sid="standard_particle">
   <std_particle_type>4</std_particle_type>
  </standard_particle>
 </particle_type>
 <particle_spin sid="particle_spin">
  <time>333.333</time>
  <time_variance>0</time_variance>
  <phase>0</phase>
  <phase_variance>0</phase_variance>
  <axis sid="axis">
   <variable_vector sid="variable_vector">
    <vector>1 1 0</vector>
    <variance>0.436332</variance>
   </variable_vector>
  </axis>
 </particle_spin>
</emitter>
```

shape

Introduction

Describes the shape of a particle emitter. This shape is a volume in space within which particles are emitted. The idea is to randomly distribute the emitted particles' initial position across the volume. Currently, there are four types of shapes supported. Box, sphere and cylinder are shapes centered around the emitter instance's node position. Target, however, is a reference to an existing geometric object in the scene. The particles will be emitted inside the geometric object's volume – the volume is situated at the geometric object's position, not at the position of the emitter instance's node.

Attributes

The <shape> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <shape> element relates to the following elements:

Parent elements	emitter
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
shape_element	The element that describes shape data. Must be exactly one of: • <box> • <sphere> • <cylinder> • <target> See main entries.	N/A	1

Details

Example

box

Introduction

Describes a box.

Attributes

The <box> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <box> element relates to the following elements:

Parent elements	shape
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<length>	Length of the box in scene distance units.	N/A	1
<width>	Width of the box in scene distance units.	N/A	1
<height>	Height of the box in scene distance units.	N/A	1

Details

Example

sphere

Introduction

Describes a sphere.

Attributes

The <sphere> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <sphere> element relates to the following elements:

Parent elements	shape
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<radius>	Radius of the sphere in scene distance units.	N/A	1

Details

Example

cylinder

Introduction

Describes a cylinder.

Attributes

The <cylinder> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <cylinder> element relates to the following elements:

Parent elements	cylinder
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<radius>	Radius of the cylinder in scene distance units.	N/A	1
<height>	Height of the cylinder in scene distance units.	N/A	1

Details

Example

length

Introduction

Describes a length in scene distance units.

Content Type

float

Attributes

The <length> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <length> element relates to the following elements:

Parent elements	box
Child	None.
Other	None.

Child Elements

Details

Example

width

Introduction

Describes a width in scene distance units.

Content Type

float

Attributes

The <width> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <width> element relates to the following elements:

Parent elements	box
Child	None.
Other	None.

Child Elements

Details

Example

height

Introduction

Describes a height in scene distance units.

Content Type

float

Attributes

The <height> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <height> element relates to the following elements:

Parent elements	box, cylinder
Child	None.
Other	None.

Child Elements

Details

Example

radius

Introduction

Describes a radius in scene distance units.

Content Type

float

Attributes

The <radius> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <radius> element relates to the following elements:

Parent elements	sphere, cylinder
Child	None.
Other	None.

Child Elements

Details

Example

emission

Introduction

Describes the emission parameters of the emitter.

Attributes

The <emission> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <emission> element relates to the following elements:

Parent elements	emitter
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<max_particles>	See main entry.	N/A	1
<birth_rate>	See main entry.	N/A	1

Details

Example

max_particles

Introduction

The maximum number of alive particles emitted that can exist at any one time.

Content Type

int

Attributes

The <max_particles> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <max_particles> element relates to the following elements:

Parent elements	emission
Child	None.
Other	None.

Child Elements

Details

Example

birth_rate

Introduction

Describes the amount of particles emitted per second.

Content Type

float

Attributes

The <birth_rate> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <birth_rate> element relates to the following elements:

Parent elements	emission
Child	None.
Other	None.

Parameters

Child Elements

Details

Example

randomization

Introduction

Describes the randomization parameters of the emitter.

Attributes

The <randomization> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <randomization> element relates to the following elements:

Parent elements	emitter
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<seed>	See main entry.	N/A	1

Details

Example

seed

Introduction

The initial seed of the random number generator associated to the emitter.

Content Type

int

Attributes

The <seed> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <seed> element relates to the following elements:

Parent elements	randomization
Child	None.
Other	None.

Child Elements

Details

Example

particle_life

Introduction

Describes the parameters affecting the lifetime of particles.

Attributes

The <particle_life> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <particle_life> element relates to the following elements:

Parent elements	emitter
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<life>	See main entry.	N/A	1
<life_variance>	See main entry.	N/A	1

Details

Example

life

Introduction

Lifetime of a particle.

Content Type

float

Attributes

The <life> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <life> element relates to the following elements:

Parent elements	particle_life
Child	None.
Other	None.

Child Elements

Details

Example

life_variance

Introduction

The variance to apply to the lifetime in time units.

Content Type

float

Attributes

The <life_variance> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <life_variance> element relates to the following elements:

Parent elements	particle_life
Child	None.
Other	None.

Child Elements

Details

Variance Equation :

PARTICLE_LIFE = LIFE + RAND_BETWEEN(-1 ,1) * LIFE_VARIANCE

Example

particle_size

Introduction

Describes the parameters affecting the size of particles.

Attributes

The <particle_size> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <particle_size> element relates to the following elements:

Parent elements	emitter
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<size>	See main entry.	N/A	1
<size_variance>	See main entry.	N/A	1
<size_grow_for>	See main entry.	N/A	1
<size_fade_for>	See main entry.	N/A	1

Details

Example

size

Introduction

Describes the size of a particle in scene distance units.

Content Type

float

Attributes

The <size> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <size> element relates to the following elements:

Parent elements	particle_size
Child	None.
Other	None.

Child Elements

Details

Example

size_variance

Introduction

The variance to apply to the size. Value between 0 and 1 representing a percentage of the size.

Content Type

float

Attributes

The <size_variance> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <size_variance> element relates to the following elements:

Parent elements	particle_size
Child	None.
Other	None.

Child Elements

Details

Variance Equation :

PARTICLE_SIZE = SIZE * (1 + RAND_BETWEEN(-1 ,1) * SIZE_VARIANCE)

Example

size_grow_for

Introduction

Duration of the size grow-in in time units.

Content Type

float

Attributes

The <size_grow_for> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <size_grow_for> element relates to the following elements:

Parent elements	particle_size
Child	None.
Other	None.

Child Elements

Details

Example

size_fade_for

Introduction

Duration of the size grow-out in time units.

Content Type

float

Attributes

The <size_fade_for> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <size_fade_for> element relates to the following elements:

Parent elements	particle_size
Child	None.
Other	None.

Child Elements

Details

Example

particle_motion

Introduction

Describes the parameters affecting the motion of particles.

Attributes

The <particle_motion> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <particle_motion> element relates to the following elements:

Parent elements	emitter
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<speed>	See main entry.	N/A	1
<speed_variance>	See main entry.	N/A	1
<direction>	See main entry.	N/A	1

Details

Example

speed

Introduction

Speed of the particle in scene distance units per second.

Content Type

float

Attributes

The <speed> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <speed> element relates to the following elements:

Parent elements	particle_motion
Child	None.
Other	None.

Child Elements

Details

Example

speed_variance

Introduction

The variance to apply to the speed. Value between 0 and 1 representing a percentage of the speed.

Content Type

float

Attributes

The <speed_variance> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <speed_variance> element relates to the following elements:

Parent elements	particle_motion
Child	None.
Other	None.

Child Elements

Details

Variance Equation :

PARTICLE_SPEED = SPEED * (1 + RAND_BETWEEN(-1 ,1) * SPEED_VARIANCE)

Example

direction

Introduction

Describes the direction of particles. The direction is either a random vector, a vector with an associated variance expressed as a percentage (vector_variance_percentage), or a node in the scene with an associated variance.

Attributes

The <direction> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <direction> element relates to the following elements:

Parent elements	particle_motion
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
direction_element	The element that describes direction data. Must be exactly one of: • <random_vector> • <variable_vector> • <direction_target> See main entries.	N/A	1

Details

Example

direction_target

Introduction

The direction target is a <target> and an associated variance.

Attributes

The <direction_target> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <direction_target> element relates to the following elements:

Parent elements	direction
Child	See Child Elements subsection.
Other	None.

Parameters

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<target>	See main entry.	N/A	1
<variance>	See main entry.	N/A	1

Details

Example

particle_type

Introduction

Describes the parameters affecting the particle type.

Attributes

The <particle_type> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <particle_type> element relates to the following elements:

Parent elements	emitter
Child	See Child Elements subsection.
Other	None.

Parameters

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
type_element	The element that describes type data. Must be exactly one of: • <standard_particle> • <meta_particle> • <instanced_geometry_particle> See main entries.	N/A	1

Details

Example

standard_particle

Introduction

Describes a standard particle.

Attributes

The <standard_particle> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <standard_particle> element relates to the following elements:

Parent elements	particle_type
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<std_particle_type>	See main entry.	N/A	1

Details

Example

std_particle_type

Introduction

Standard particles can come in a number of shapes. This element contains an integer value corresponding to one of these shapes.

Content Type

int

Attributes

The <std_particle_type> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <std_particle_type> element relates to the following elements:

Parent elements	standard_particle
Child	None.
Other	None.

Child Elements

Details

Possible Values :

Value	Description
0	Triangle
1	Cube
2	Special
3	Facing (billboard)
4	Constant
5	Tetrahedron
6	Six-pointed star
7	Sphere

Example

meta_particle

Introduction

Describes a meta particle.

Attributes

The <meta_particle> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <meta_particle> element relates to the following elements:

Parent elements	particle_type
Child	None.
Other	None.

Parameters

Child Elements

Details

Example

instanced_geometry_particle

Introduction

Describes an instanced geometry particle.

Attributes

The <instanced_geometry_particle> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <instanced_geometry_particle> element relates to the following elements:

Parent elements	particle_type
Child	See Child Elements subsection.
Other	None.

Parameters

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<target>	See main entry.	N/A	1
<use_subtree>	See main entry.	N/A	1

Details

Example

use_subtree

Introduction

Describes whether to use the descendants of the instanced geometry node to render each particle.

Content Type

bool

Attributes

The <instanced_geometry_particle> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <instanced_geometry_particle> element relates to the following elements:

Parent elements	instanced_geometry_particle
Child	None.
Other	None.

Parameters

Child Elements

Details

Example

particle_animation_offset

Introduction

Describes the particle animation offset.

Attributes

The <particle_animation_offset> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <particle_animation_offset> element relates to the following elements:

Parent elements	emitter
Child	See Child Elements subsection.
Other	None.

Parameters

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<method>	See main entry.	N/A	1
<offset>	See main entry.	N/A	1
<offset_variance>	See main entry.	N/A	1

Details

Example

method

Introduction

Describes the method used to offset the animation.

Content Type

string

Attributes

The <method> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <method> element relates to the following elements:

Parent elements	particle_animation_offset
Child	None.
Other	None.

Parameters

Child Elements

Details

Possible Values :

Value	Description
NONE	No offset applied.
BIRTH	Offset applied from birth.

Example

offset

Introduction

Describes the offset in time units to apply to the animation.

Content Type

float

Attributes

The <offset> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <offset> element relates to the following elements:

Parent elements	particle_animation_offset
Child	None.
Other	None.

Parameters

Child Elements

Details

Example

offset_variance

Introduction

Describes the variance to apply to the offset. Percentage value.

Content Type

float

Attributes

The <offset_variance> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <offset_variance> element relates to the following elements:

Parent elements	particle_animation_offset
Child	None.
Other	None.

Parameters

Child Elements

Details

Example

particle_animation_scaling

Introduction

Describes the particle animation offset.

Attributes

The <particle_animation_scaling> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <particle_animation_scaling> element relates to the following elements:

Parent elements	emitter
Child	See Child Elements subsection.
Other	None.

Parameters

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<method>	See main entry.	N/A	1
<src_length>	See main entry.	N/A	1
<src_variance>	See main entry.	N/A	1
<dst_multiplier>	See main entry.	N/A	1
<dst_variance>	See main entry.	N/A	1

Details

Example

method

Introduction

Describes the method used to scale the animation.

Content Type

string

Attributes

The <method> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <method> element relates to the following elements:

Parent elements	particle_animation_scaling
Child	None.
Other	None.

Parameters

Child Elements

Details

Possible Values :

Value	Description
NONE	No scaling applied.
LIFE	Scales the source animation to the life of the particle.

Example

src_length

Introduction

Describes the animation source length in time units.

Content Type

float

Attributes

The <src_length> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <src_length> element relates to the following elements:

Parent elements	particle_animation_scaling
Child	None.
Other	None.

Parameters

Child Elements

Details

Example

src_variance

Introduction

Describes the variance to apply to the animation scaling source length. Percentage value.

Content Type

float

Attributes

The <src_variance> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <src_variance> element relates to the following elements:

Parent elements	particle_animation_scaling
Child	None.
Other	None.

Parameters

Child Elements

Details

Example

dst_multiplier

Introduction

Describes the multiplier to apply the animation scale destination length.

Content Type

float

Attributes

The <dst_multiplier> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <dst_multiplier> element relates to the following elements:

Parent elements	particle_animation_scaling
Child	None.
Other	None.

Parameters

Child Elements

Details

For example, scaling an animation to a particle whose life is 2 with a multipler of 2 will scale the animation to a length a 4.

Example

dst_variance

Introduction

Describes the per particle variance to apply to animation scaling destination length multiplier. Percentage value.

Content Type

float

Attributes

The <dst_variance> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <dst_variance> element relates to the following elements:

Parent elements	particle_animation_scaling
Child	None.
Other	None.

Parameters

Child Elements

Details

Example

particle_spin

Introduction

Describes the parameters affecting the spin of particles. The spin is a rotation animation around an axis passing through the particle's center. The particle spin is defined by three main parameters : time, phase and axis. The time is the rotation period – the time it takes to rotate 360 degrees around the axis. Phase is an offset to apply to the spin, in degrees. The axis is self-explanatory.

Attributes

The <particle_spin> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <particle_spin> element relates to the following elements:

Parent elements	emitter
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<time>	See main entry.	N/A	1
<time_variance>	See main entry.	N/A	1
<phase>	See main entry.	N/A	1
<phase_variance>	See main entry.	N/A	1
<axis>	See main entry.	N/A	1

Details

Example

time

Introduction

Spin period of the particle in time units.

Content Type

float

Attributes

The <time> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <time> element relates to the following elements:

Parent elements	particle_spin
Child	None.
Other	None.

Child Elements

Details

Example

time_variance

Introduction

The variance to apply to the spin period. Value between 0 and 1 representing a percentage of the spin period.

Content Type

float

Attributes

The <time_variance> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <time_variance> element relates to the following elements:

Parent elements	particle_spin
Child	None.
Other	None.

Child Elements

Details

Variance Equation :

PARTICLE_SPIN_PERIOD = TIME * (1 + RAND_BETWEEN(-1 ,1) * TIME_VARIANCE)

Example

phase

Introduction

Spin phase in degrees.

Content Type

float

Attributes

The <phase> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <phase> element relates to the following elements:

Parent elements	particle_spin
Child	None.
Other	None.

Child Elements

Details

Example

phase_variance

Introduction

The variance to apply to the spin phase. Value between 0 and 1 representing a percentage of the spin sphase.

Content Type

float

Attributes

The <phase_variance> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <phase_variance> element relates to the following elements:

Parent elements	particle_spin
Child	None.
Other	None.

Child Elements

Details

Variance Equation :

PARTICLE_SPIN_PHASE = PHASE * (1 + RAND_BETWEEN(-1 ,1) * PHASE_VARIANCE)

Example

axis

Introduction

Describes the axis around which particles spin.

Attributes

The <axis> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <axis> element relates to the following elements:

Parent elements	particle_spin
Child	See Child Elements subsection.
Other	None.

Parameters

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
axis_element	The element that describes type data. Must be exactly one of: • <random_vector> • <variable_vector> • <travel_or_mblur_dir> See main entries.	N/A	1

Details

Example

travel_or_mblur_dir

Introduction

TODO.

Attributes

The <travel_or_mblur_dir> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <travel_or_mblur_dir> element relates to the following elements:

Parent elements	axis
Child	See Child Elements subsection.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<stretch>	See main entry.	N/A	1

Details

Example

stretch

Introduction

TODO.

Content Type

float

Attributes

The <stretch> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <stretch> element relates to the following elements:

Parent elements	travel_or_mblur_dir
Child	None.
Other	None.

Child Elements

Details

Example

random_vector

Introduction

Describes a random vector.

Attributes

The <random_vector> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <random_vector> element relates to the following elements:

Parent elements	direction, axis
Child	None.
Other	None.

Child Elements

Details

Example

variable_vector

Introduction

Describes a vector with an associated variance.

Attributes

The <variable_vector> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <variable_vector> element relates to the following elements:

Parent elements	direction, axis
Child	None.
Other	None.

Child Elements

Child elements must appear in the following order if present:

Name/example	Description	Default	Occurrences
<vector>	See main entry.	N/A	1
<variance>	See main entry.	N/A	1

Details

Example

vector

Introduction

Describes a vector.

Content Type

float3

Attributes

The <vector> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <vector> element relates to the following elements:

Parent elements	variable_vector
Child	See Child Elements subsection.
Other	None.

Child Elements

Details

Example

quantity

Introduction

Describes a quantity of something.

Content Type

float

Attributes

The <quantity> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <quantity> element relates to the following elements:

Parent elements	None.
Child	None.
Other	None.

Child Elements

Details

Example

target

Introduction

Describes a target using the Collada targeting system syntax (see Collada spec). If element is empty, consider the target to be invalid.

Content Type

URI

Attributes

The <target> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <target> element relates to the following elements:

Parent elements	shape, direction_target, instanced_geometry_particle
Child	None.
Other	None.

Child Elements

Details

Example

variance

Introduction

Describes a variance whose interpretation depends on the context.

Content Type

float

Attributes

The <variance> element has the following attributes:

sid	**xs:NCName**	A text string value containing the subidentifier of this element. This value must be unique within the scope of the parent element.

Related Elements

The <variance> element relates to the following elements:

Parent elements	direction_target, variable_vector
Child	None.
Other	None.

Child Elements

Details

Example